

UNIVERSITY OF CAMBRIDGE

NATURAL SCIENCES TRIPOS Part II

Thursday 4th June 2026

09:00 to 11:00

PHYSICS

PHYSICAL SCIENCES: HALF SUBJECT PHYSICS

ADVANCED AND QUANTUM OPTICS — Paper 2

*Candidates offering this paper should attempt **all six** questions (four from Section A and two from Section B).*

The approximate number of marks allocated to each question or part of a question is indicated in the right margin. This paper contains 4 sides, including this coversheet, and is accompanied by a handbook giving values of constants and containing mathematical formulae which you may quote without proof.

STATIONERY REQUIREMENTS

2 × 8-page answer booklet

Rough workpad

Yellow master coversheet

SPECIAL REQUIREMENTS

Mathematical Formulae Handbook

Approved calculator allowed

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the Invigilator.

SECTION A

Attempt **all four** questions in this Section. Answers should be concise and relevant formulae may be assumed without proof.

1 A monochromatic light source has a Gaussian spectral profile with a full width at half maximum (FWHM) linewidth $\Delta\nu = 1.5$ GHz. Determine the coherence time τ_c , the coherence length l_c , and the value of $|g(\tau)|$ at the delay $\tau = 200$ ps. You may use the result that for a Gaussian profile, $\tau_c = \sqrt{2 \ln 2 / \pi} / \Delta\nu$.

2 The ray-transfer matrix for propagation through a distance d of free space is $\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}$, and for a thin lens of focal length f it is $\begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix}$. Calculate the overall ray-transfer matrix for a $4f$ imaging system consisting of two lenses of focal lengths $f_1 = 10$ cm and $f_2 = 20$ cm separated by $f_1 + f_2$, with the object located at distance f_1 before the first lens and the image at distance f_2 after the second lens. Hence determine the lateral magnification.

3 An acousto-optic modulator uses a fused-silica medium ($n = 1.46$, $v_s = 6$ km/s) driven at an acoustic frequency $f = 50$ MHz. A He-Ne laser beam of wavelength $\lambda_0 = 633$ nm is incident on the device. Calculate the Bragg angle θ_B , the acoustic wavelength Λ , and the frequency shift imparted to the first-order diffracted beam.

4 A step-index optical fibre has a core refractive index $n_1 = 1.462$ and a cladding refractive index $n_2 = 1.457$. Calculate the numerical aperture and the critical angle for total internal reflection at the core-cladding interface. If the core diameter is $8.0 \mu\text{m}$, determine whether the fibre is single-mode at $\lambda_0 = 1.55 \mu\text{m}$.

SECTION B

Attempt **both** questions from this Section.

5 This question concerns optical resonators and laser oscillation.

- (a) A planar Fabry–Perot resonator of length d and refractive index n has mirrors with amplitude reflectances whose product gives an intensity round-trip attenuation factor $|r|^2 < 1$. Starting from the superposition of multiply-reflected waves, show that the intracavity intensity is

$$I = \frac{I_{\max}}{1 + (2\mathcal{F}/\pi)^2 \sin^2(\phi/2)},$$

where $\phi = k \cdot 2d$ is the round-trip phase, and identify the finesse $\mathcal{F}$ in terms of $|r|$.

- (b) Hence derive expressions for the resonance frequencies ν_q , the free spectral range ν_F , and the spectral width $\delta\nu$ of the resonator modes.
- (c) Now consider a spherical-mirror resonator consisting of two concave mirrors of radii R_1 and R_2 separated by a distance d . State the stability condition in terms of the parameters $g_1 = 1 - d/R_1$ and $g_2 = 1 - d/R_2$. Sketch a stability diagram in the (g_1, g_2) plane, indicating the regions corresponding to planar, confocal, concentric, and symmetric resonators.
- (d) A Nd:YVO₄ laser operates at $\lambda = 1064$ nm in a resonator of length $d = 15$ cm with mirrors of reflectance $\mathcal{R}_1 = 1.00$ and $\mathcal{R}_2 = 0.95$. The gain medium has a Doppler-broadened linewidth $\Delta\nu_D = 300$ GHz. Calculate the number of longitudinal modes that lie within the gain bandwidth, and the photon lifetime τ_p of the resonator. If the peak small-signal gain coefficient is $\gamma_0 = 0.20$ m⁻¹, determine whether the laser reaches threshold.

6 This question concerns nonlinear optics and electro-optic effects.

- (a) Starting from the anharmonic oscillator model with potential $V = \frac{1}{2}k_0x^2 + \frac{1}{3}k_1x^3$, explain qualitatively why the polarisation of a non-centrosymmetric medium contains terms at frequencies 2ω and 0 when driven by a monochromatic field $E(t) = E_0 \cos(\omega t)$. Explain why $k_1 = 0$ in a centrosymmetric medium.
- (b) Two plane waves at frequencies ω_1 and ω_2 propagate collinearly in a $\chi^{(2)}$ medium. Write down the five distinct frequency components present in the second-order nonlinear polarisation $P^{(2)}$, and identify the processes of second-harmonic generation, sum-frequency generation, difference-frequency generation, and optical rectification.
- (c) Explain the optical Kerr effect, giving the expression $n(I) = n_0 + n_2I$. A short optical pulse propagates through a Kerr medium. Starting from the phase $\varphi(t) = \omega_0 t - k_0 z[n_0 + n_2I(t)]$, derive an expression for the instantaneous frequency $\omega(t)$ and explain the phenomenon of self-phase modulation. What determines whether the resulting chirp leads to pulse compression or broadening?

- (d) Explain how, in a medium with anomalous group-velocity dispersion and a positive Kerr coefficient $n_2 > 0$, these two effects can balance to produce a temporal soliton. Write down the nonlinear Schrödinger equation governing the pulse envelope and state the soliton solution.

END OF PAPER